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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' GRADE Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE & M.TECH - CSE

20CS2E11 - BIGDATA TECHNOLOGIES

Answer ALL questions

Duration: 2 Hrs 30 min

Maximum: 70 Marks

PART - A

(5x 1 = 5 Marks)

1. **CO1** XML, JSON, and CSV files are examples of what type of data?
a) Structured b) Semi-Structured c) Unstructured d) Random Data
2. **CO2** Which command-line interface does Hadoop primarily use?
a) Hadoop CLI b) Bash only c) Hadoop Streaming d) Hadoop Shell
3. **CO2** Data locality in Hadoop is primarily managed by _____.
a) Resource Manager b) NameNode
c) DataNode d) JobTracker / YARN scheduler
4. **CO3** The Metastore in Hive stores _____.
a) HDFS blocks b) Data schemas and metadata
c) User accounts d) MapReduce logs
5. **CO3** Chukwa is built on top of which system?
a) Hive and Pig b) Spark and YARN
c) HDFS and MapReduce d) HBase and Flume

PART - B

(5x 3 = 15 Marks)

6. **CO1** What is the CAP Theorem in Big Data? Explain its components with examples.
7. **CO2** List and briefly explain the core components of Hadoop.
8. **CO2** What is parallel efficiency in the context of MapReduce?

9. **CO3** How does Mahout use Hadoop for machine learning?
10. **CO3** How do Chukwa and Flume differ?

PART - C

(5 x 10 = 50 Marks)

Q.No: 11 Compulsory Question

Q.No: 12 to 17 Answer any FOUR Questions

Compulsory Question:

11. **CO1** i) Explain the Hadoop ecosystem and its components. How does it support Big Data analytics? (5)
- CO1** ii) Why can't a distributed system achieve Consistency, Availability, and Partition Tolerance all at the same time? Explain with an example. (5)
12. **CO1** i) Explain the advantages and limitations of Big Data analytics compared to traditional Business Intelligence (BI). (5)
- CO1** ii) What are the main stages of a Data Science workflow? (5)
13. **CO2** i) A retail company has large sales, inventory, and customer feedback data from multiple stores. How can Hadoop help manage and analyze this data? (5)
- CO2** ii) Explain the role of NameNode and DataNode in HDFS. Why are they important. (5)
14. **CO2** i) Explain the concept of parallel efficiency in MapReduce. How does Hadoop achieve high parallel efficiency? (5)
- CO2** ii) A company wants to analyze tweets and posts to understand customer sentiment in real-time. How can Big Data tools help? (5)
15. **CO3** i) A social media company needs to store user profiles and access them in real-time. How can HBase aid? (5)
- CO3** ii) A telecom company collects millions of call detail records (CDRs) daily. The data is messy and needs cleaning, filtering, and aggregation before analysis. How can Apache Pig help? (5)

16. **CO3** i) A company collects millions of log messages from its web servers daily. (5)
The logs need to be transferred efficiently to Hadoop for analysis while maintaining schema consistency. How can Apache Avro assist?
- CO3** ii) What is Apache Oozie and how does it help in managing workflows in Hadoop? (5)
17. **CO3** i) A distributed system has multiple HBase region servers that must stay in sync to avoid data inconsistency. Occasionally, a server crashes. How can ZooKeeper help maintain consistency? (5)
- CO2** ii) A retail company needs total sales per product across hundreds of stores. (5)
How can MapReduce be applied?
